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Fast and accurate multiple-protein-sequence alignment at scale with FAMSA2

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We introduce FAMSA2, an algorithm that produces high-accuracy multiple-protein-sequence alignments at high speed. Across structural, phylogenetic and functional benchmarks, FAMSA2 matches or exceeds the accuracy of state-of-the-art tools while running 400 times faster on average than existing aligners. By combining progressive alignment with medoid clustering-based guide tree construction and a dissimilarity measure derived from the longest common subsequence, it enables scalable alignment of massive protein families.

The output of sequencing has increased billionfold in less than 30 years[1], fueling initiatives such as the Earth BioGenome Project[2] and MGnify[3], which aim to catalog all genomic information on the planet. As a result, protein sequence databases will soon contain billions of entries. To keep pace, sequence similarity search tools able to process this amount of data have been developed (DIAMOND[4] and MMseqs2 (ref. 5)), enabling the discovery of hundreds of thousands of novel protein families[6]. In parallel, the AlphaFold revolution[7] and the release of over 200 million protein structures[8] have driven the development of structure-based clustering tools (that is, FoldSeek[9]), further enriching protein family diversity. However, multiple-sequence alignment (MSA) tools—crucial for inferring evolutionary relationships, protein structure and phylogeny—lag behind the rapidly growing protein data. The most accurate aligners for large datasets, Muscle5 (ref. 10) and T-Coffee's regressive method[11], scale poorly beyond thousands of sequences, with accuracy degrading as dataset size and noise increase[12]. As protein families grow toward millions of members, alignment approaches that overcome these limitations are needed.

We address these issues with FAMSA2, a progressive MSA algorithm (Fig. 1a) that aligns millions of sequences in minutes, efficiently using modern hardware. FAMSA2 reaches or exceeds the accuracy of state-of-the-art tools, exhibits greater throughput than the fastest existing methods, shows superior robustness to nonhomologous contamination and provides usability features not offered by other approaches or its predecessor[13] (Fig. 1b). FAMSA2 introduces a stochastic medoid tree algorithm that removes practical barriers to large-scale alignment by generating stable guide trees in $O(MN\log N)$ time—where N is the number of sequences and M the maximum subtree size—with accuracy comparable to default $O(N^2)$-time single-linkage trees (Fig. 1c). This guide tree approximation is supported by algorithmic advances that improve both alignment accuracy and throughput, including a longest common subsequence (LCS)-based dissimilarity measure resilient to sequence divergence, length-based prefiltering of pairwise comparisons, multilevel parallelism at LCS calculation (Fig. 1d) and progressive alignment stage, parallel construction of single-linkage guide trees in linear space using Prim's algorithm (Fig. 1e) and dynamic refinement of partial alignments.

We evaluated FAMSA2 across benchmarks, assessing structural, phylogenetic and functional alignment accuracy. The structural accuracy was measured using an updated extHomFam[13], the largest structure-based benchmark, containing 390 protein families with up to 3 million sequences. Each family includes a subset of sequences with curated, structurally based reference alignments, used to measure the percentage of correctly aligned residue pairs (sum-of-pairs score; SP) and columns (total-column score; TC) (Methods). Muscle5, T-Coffee, UPP and Clustal Omega required 6–37 days to process this benchmark and, with the exception of UPP, failed at the largest ABC transporter family (Fig. 2a and Supplementary Table 1). In contrast, FAMSA2 produced the most accurate alignments in 15 h (SP = 79.6, TC = 61.8), while FAMSA2 with medoid trees required only 47 min to deliver results comparable to Muscle5 (>400× slower) and only slightly inferior to T-Coffee (>1,130× slower). Moreover, this fast mode of FAMSA2 surpassed Clustal Omega, MAFFT and Kalign by more than 15 points in SP and TC (Fig. 2a), showing that medoid tree properties (Extended Data Fig. 1) translate to superior accuracy. The advantages of FAMSA2 in accuracy and throughput became more pronounced as protein family size increased

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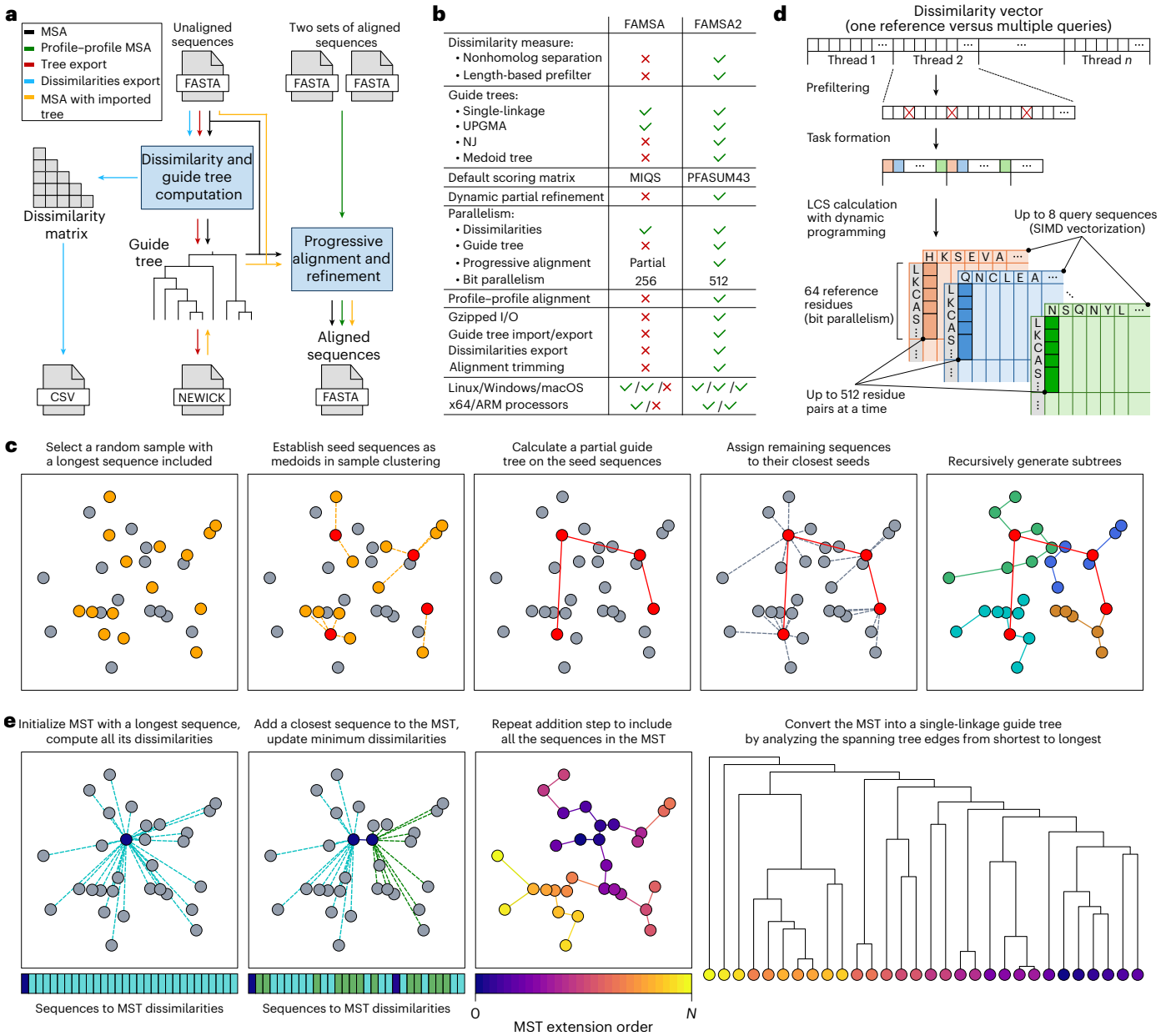

Fig. 1 | FAMSA2 algorithms and features. a, Usage scenarios presented as data streams. **b**, Major differences between FAMSA and FAMSA2. **c**, Medoid tree algorithm for fast guide tree approximation with reduced stochasticity in seed selection and improved stability over fully random algorithms like PartTree. **d**, Parallel dissimilarity calculation between the currently analyzed sequence (reference) and all other sequences in a set (queries) during guide tree construction. **e**, Single-linkage guide tree construction with linear space using Prim's MST algorithm. I/O, input/output; SIMD, single instruction, multiple data.

(Fig. 2b and Supplementary Table 1). By combining guide tree construction and alignment generation components from different tools, we found that the FAMSA2 guide trees and alignment strategy contribute additively to the accuracy (Extended Data Fig. 2).

To further assess accuracy scaling, we analyzed the largest extHomFam families subsampled to contain from 100 to 100,000 homologs (Fig. 2c and Methods). FAMSA2 ranked first in accuracy for families with ≥5,000 sequences, outrunning Muscle5 from $N \geq 700$ and T-Coffee from $N \geq 5{,}000$. It also showed the most stable performance across all family sizes, with an SP score decrease of only 3.6 points, compared to 9.2-point and 6.7-point declines for Muscle5 and T-Coffee, respectively (Fig. 2c and Supplementary Table 2). FAMSA2's medoid trees had comparable stability and accuracy to Muscle5, while the other tools—MAFFT, Clustal Omega and Kalign—showed lower accuracy, with SP score declines exceeding 20 points.

To investigate alignment robustness to contamination by unrelated sequences, we generated datasets containing an increasing number of mixed homologous and nonhomologous sequences, scaling up to over 12 million sequences (Fig. 2d and Methods). Only FAMSA and FAMSA2 completed all datasets, as the others failed because of memory or time constraints (Fig. 2d and Supplementary Table 3). FAMSA2 achieved the highest SP and TC scores across 12 of 14 datasets and, unlike other tools, improved its accuracy as dataset size increased. This resulted from the modified dissimilarity measure, which effectively clusters related sequences (Extended Data Fig. 3). FAMSA2's medoid tree ranked third in SP and TC stability and overall accuracy, following FAMSA and FAMSA2 (Fig. 2d).

To address the potential limitations of the extHomFam benchmark (that is, its reliance on a small number of reference structures), we aligned ~1,100 protein families (1,000–680,000 sequences) from

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a

	SP (%)	TC (%)	Total time	Memory (GB)
Kalign	61.9	43.2	1 h 8 min	140.2
MAFFT DPPartTree	62.9	44.3	8 days 10 h 57 min	1067.7
Clustal Omega	63.8	46.9	7 days 2 h 44 min	144.4
UPP	70.4	53.6	6 days 2 h 44 min	158.8
Muscle5	75.7	59.4	13 days 3 h 39 min	225.2
T-Coffee regressive	78.1	61.3	36 days 22 h 58 min	410.1
FAMSA	78.1	60.1	1 day 7 h 57 min	46.9
FAMSA2	79.6	61.8	14 h 57 min	23.9
FAMSA2 medoid	76.5	58.5	47 min	17.7

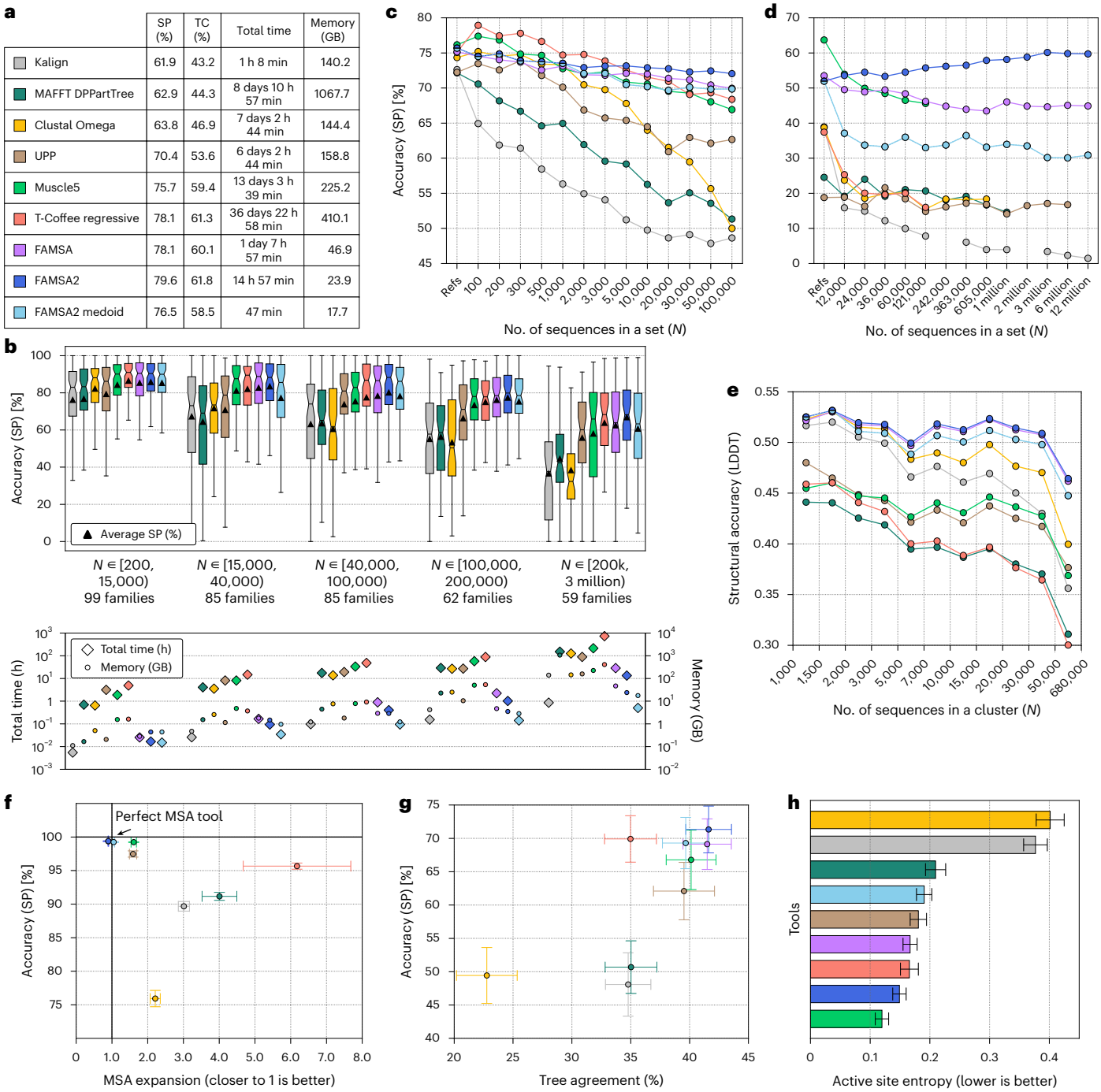

Fig. 2 | Benchmarking FAMSA2 across diverse datasets. a, Average accuracy, total runtime and peak memory usage on 390 extHomFam protein families. Accuracy is shown as SP and TC scores. **b**, Accuracy, runtime and peak memory across five extHomFam family size groups. Box plots legend: central line, median; notches, 95% confidence interval (CI) of the median; triangle, mean; box, lower (Q1) and upper (Q3) quartiles; whiskers, lower (Q1 − 1.5× IQR) and upper (Q3 + 1.5× IQR) outlier bounds, where IQR is the interquartile range (Q3 − Q1). **c**, Average SP scores for 121 largest extHomFam families subsampled to contain 100–100,000 homologs. **d**, Average SP scores for mixed homologous and nonhomologous datasets. **e**, Structural MSA accuracy as bootstrapped LDDT score on clusters of increasing sizes sampled from AlphaFold Database. **f**, Average phylogenetic accuracy (MSA expansion; the number of columns relative to the reference MSA) and structural accuracy (SP score) on 1,860 simulated MSAs. Error bars indicate the 95% CI. **g**, Average phylogenetic accuracy (tree agreement) and structural accuracy (SP score) on 113 extHomFam families sampled to 100,000 sequences. Error bars indicate the 95% CI. **h**, Accuracy on 772 enzyme families with 1,376 known active sites measured as an average entropy of active-site residue distribution across alignment columns. Error bars indicate the 95% CI. A summary of these benchmark results is provided in Extended Data Table 1.

AlphaFold Database clusters[8] and evaluated the resulting MSAs using the local distance difference test (LDDT) score[14]—a reference-free accuracy measure based on the local structures superposition (0–1; Methods). Both FAMSA2 variants achieved the highest average LDDT scores (0.50–0.51) (Extended Data Table 1 and Supplementary Table 4) and had the smallest decline in accuracy when increasing dataset sizes (Fig. 2e). Notably, Clustal Omega (0.48) and Kalign (0.46), which ranked lower in the previous benchmarks, surpassed Muscle5 (0.43) and T-Coffee (0.40) in this structural assessment. The medoid version of FAMSA2 required 227-fold less time than Clustal Omega (Supplementary Table 5), the

next most accurate method, while Kalign, the only algorithm with comparable execution times, delivered alignments of notably lower quality. Overall, FAMSA2 processed the entire database of ~25,000 families in half a day compared to an estimated ~59 days for Clustal Omega.

Given that structural accuracy is only a proxy of evolutionary correctness, we investigated the capacity of alignments to retain phylogenetic information using structure-free experimental protocols. First, we simulated 1,860 protein families along known trees (Methods), yielding reference MSAs grounded in evolutionary history. Both FAMSA2 variants achieved the highest SP and TC scores (Fig. 2f and Supplementary Table 6) with near-ideal expansion (the ratio of the number of columns in the tool's MSA to those in the reference MSA) (Fig. 2f). Next, we assessed phylogenetic signal on real protein families. In the absence of reference phylogenies, we compared maximum likelihood (ML) trees, often considered a ground truth in phylogenetic comparison, and distance-based (minimum evolution; ME) trees reconstructed from the extHomFam protein families, using topological agreement as a proxy for phylogenetic accuracy (Methods and Supplementary Table 7). FAMSA2 achieved the highest tree agreement score of 41.6%, followed closely by FAMSA (41.5%) and Muscle5 (40.1%) (Fig. 2g). The FAMSA2 medoid tree ranked next with 39.7%, slightly surpassing UPP (39.5%) and outperforming the other tools by 4–17 percentage points. Across families, Pearson correlations between the gap fraction and ME–ML agreement were positive (median: 0.651 ± 0.290, 110/115 positive Pearson values; Supplementary Table 8), indicating that lower agreement is not associated with higher indel density. While high ME–ML agreement alone does not demonstrate improved phylogenetic accuracy, it indicates that FAMSA2 produces MSAs that are potentially better substrates for fast heuristic tree-building methods such as ME, which are often necessary for large-scale analyses.

Our final experiment focused on functional alignment, as structure-based and phylogeny-based benchmarks may be confounded by convergent evolution when used for functional modeling. We assessed the preservation of 1,376 active sites across 772 Pfam enzyme families by measuring the entropy of the distribution of active-site residues across alignment columns where lower entropy indicates better preservation (Methods and Supplementary Table 9). FAMSA2 achieved an entropy of 0.15 (second after Muscle5; 0.12) (Fig. 2g) retaining 97% of active-site residues on average in the most populated columns (Extended Data Fig. 4). The medoid-based variant of FAMSA2 also produced accurate results (0.19), slightly surpassing MAFFT (0.21) and outperforming Kalign (0.38) and Clustal Omega (0.40), which both spread active sites across more columns (<95% of residues in the first column).

In conclusion, FAMSA2 combines high accuracy with scalability, enabling efficient exploitation of current and future multicore architectures (Extended Data Fig. 5). It excels in aligning massive protein families—such as the ~3-million-sequence ABC transporter family in 5 min using 18 GB of RAM—and processing large collections, including ~22,000 Pfam-A families (~62 million sequences) in 8 h and 100,000 recently discovered protein families (~20 million sequences)[6] in ~5 h. The medoid tree algorithm provides further evidence on the suitability of non-biological trees for MSA reconstruction[15,16]. By rendering high-quality alignments in $O(MN \log N)$ time (complexity assumptions validated experimentally in Extended Data Fig. 6), it brings renewed long-term sustainability to progressive alignment. With FAMSA2, MSAs are now able to scale with the databases of increasing size and family discovery tools, ready to face the challenge of sequencing entire ecosystems.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41587-026-03095-3.

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Methods

FAMSA2 overview

FAMSA2 retains the two-stage progressive alignment approach of its predecessor. In the first stage, the guide tree is constructed using either our new medoid tree algorithm or a standard single-linkage clustering (other tree algorithms such as the unweighted pair group method with arithmetic mean (UPGMA) or neighbor joining (NJ) are available as options). In the second stage, the sequences are aligned into profiles (intermediate alignments) following the incorporation order indicated by the guide tree.

Beyond full MSA computation, FAMSA2 supports the following workflows: (1) guide tree generation; (2) alignment using an external guide tree; (3) dissimilarity or identity matrix calculation; and (4) alignment of two input profiles (Fig. 1a).

Medoid tree. Medoid tree is a stochastic algorithm for guide tree approximation characterized by improved scalability compared to Prim's algorithm used for the default single-linkage trees. Medoid trees require $O(M\,N\log N)$ time (where M is the algorithm parameter representing maximum subtree size), which is a notable advance over Prim's $O(N^2)$ for large sequence sets.

The medoid tree algorithm is inspired by MAFFT PartTree[17,18] and addresses its main issue of substantial randomness. PartTree performs a divisive sequence clustering through the following steps:

1. Assign all N sequences to the initial cluster.
2. Select $k < N$ seed sequences (default: $k = 50$) consisting of the longest sequence, the most dissimilar sequence to it and $k - 2$ randomly selected sequences.
3. Build a tree on seed sequences using UPGMA.
4. If there are nonseed sequences remaining:
   a. Assign nonseed sequences to their closest seeds to produce subclusters.
   b. Recursively apply the clustering procedure in each subcluster.
   c. Combine the resulting child trees with the current tree.
5. Return the tree.

While the default PartTree uses a fast, 6-mer-based dissimilarity measure, there are also modes performing more accurate alignment-based sequence comparisons: FastaPartTree and DPPartTree.

The random selection of seed sequences in PartTree results in the large instability of resulting trees; the topologies are sensitive to any variation, including reordering of the input sequences. The FAMSA2 medoid trees algorithm reduces stochasticity at the seed selection stage by performing a partition around medoids (PAM) clustering and using the resulting cluster centers (medoids) as seed sequences. The remaining clustering steps are the same as in PartTree, with the exception that single-linkage subtrees are used by default instead of UPGMA (UPGMA and NJ subtrees are also supported).

PAM[19] is an optimization procedure that selects k medoids (representatives) from N points by minimizing the cost, defined as the sum of dissimilarities between nonmedoids and their closest medoids. As the algorithm takes all pairwise distances as input, the preparation stage requires $O(N^2)$ time (assuming a dissimilarity can be computed in constant time). The canonical PAM algorithm is deterministic and consists of two steps:

1. BUILD: Selects initial medoids in $O(N^2k)$ time.
2. SWAP: Iteratively selects the best possible medoid/nonmedoid swaps until no improvement is possible or the maximum number of iterations is reached, resulting in the total time of $O(ik(N-k)^2)$, where i is the number of iterations.

These time complexities are prohibitive for large N, thus resulting in the several optimizations of PAM that improve execution times at the cost of introducing stochasticity from sampling:

- CLARA[19]: Runs PAM on a random subsample of size M ($2k < M < N$). This results in a time complexity of BUILD and SWAP steps together of $O(ikM^2)$.
- CLARANS[20]: Replaces BUILD with a random initial medoid selection ($O(k)$ time) and restricts the SWAP step to a random subset of $k(M-k)$ possible swaps (maxneighbor parameter); maxneighbor is often set relative to the number of possible swaps resulting in $O(kM^2)$ clustering time.
- FastCLARANS[17]: Eliminates $O(k)$ redundant computations in SWAP, further reducing the time complexity of the clustering algorithm to $O(M^2)$.

The medoid tree algorithm recursively divides sequences into k clusters until they contain no more than M sequences according to the following steps:

1. Initialization. Assign all N sequences to the initial cluster.
2. Recursive clustering.
   If cluster size > M (default: $M = 2{,}000$):
   a. Seed selection. Determine k seeds (default: $k = 100$) using medoid clustering:
      i. Draw a random sample of M sequences from the cluster (including the longest sequence).
      ii. Perform FastCLARANS clustering (maxneighbor = $0.1 \times k(M-k)$). The longest sequence is fixed as one of the medoids.
      iii. Return the resulting medoids as seeds.
   b. Internal tree construction. Construct a single-linkage tree T on the seed sequences using Prim's algorithm.
   c. Assignment. Assign each nonseed sequence to its closest seed, forming k subclusters.
   d. Recursion. Apply clustering procedure to each subcluster (go to step 2) to obtain child trees $T_1, \ldots, T_k$.
   e. Tree extension: Extend T with the child trees $T_1, \ldots, T_k$. Otherwise (cluster size ≤ M):
   f. Bottom tree construction. Compute a single-linkage tree T on all sequences in the cluster using Prim's algorithm.
3. Termination. Return T to the higher recursion level (or as the final guide tree on the top level).

Running the medoid tree on ≤M sequences is equivalent to FAMSA2's standard mode. To maximize the computational performance of the algorithm, the recursive clustering of subclusters at the top recursion level is parallelized. Deeper recursion levels are processed sequentially to limit thread proliferation during the divide-and-conquer process.

The time complexity of medoid tree algorithm is $O(M\,N\log N)$ (complexity analysis in Supplementary Information).

Single-linkage guide tree calculation using Prim's algorithm. FAMSA2 uses Prim's minimum spanning tree (MST) algorithm[21] for single-linkage guide tree calculation. Prim's algorithm partitions sequences into two sets: M (sequences already included in an MST) and R (sequences not yet added to the MST). For each sequence in R, the smallest dissimilarity value to any of the sequences from M is tracked. At each step, the sequence r from R with the minimum dissimilarity is selected (parallelized across threads). The selected sequence r is moved from R to M and the dissimilarity values between r and all remaining sequences in R are calculated (this step is also parallelized). This procedure repeats until R is empty. Prim's algorithm is highly parallelizable and requires only linear space with the number of sequences (N), making it well suited for large datasets.

Dissimilarity measure. For each pair of sequences A and B, the dissimilarity is calculated as: $\mathrm{indel}(A,B)^{0.75}/\mathrm{LLCS}(A,B)$, where

- $\mathrm{LLCS}(A,B)$ is the length of an LCS for A and B;
- $\mathrm{indel}(A,B)$ is the minimal number of single-symbol insertions or deletions necessary to transform A into B. It can be

easily derived from LLCS: indel(A,B) = $|A| + |B| - 2$ LLCS(A,B). The exponent in the dissimilarity measure was adjusted from 1.0 in FAMSA to 0.75 in FAMSA2 on the basis of evaluations using external reference datasets that were not part of the main benchmarks (Supplementary Table 10). This choice was further supported by the analyses showing improved robustness to the sequence divergence (Fig. 2d and Extended Data Fig. 3).

LLCS is calculated using the bit-parallel approach that compares 64 residue pairs simultaneously. FAMSA2 supports CPU SIMD extensions including AVX, AVX2, AVX512 (novel in FAMSA2) and NEON (novel in FAMSA2). For example, with AVX512, a single thread can perform eight 64-symbol comparisons concurrently. Given that many threads can operate at the same time, on a 64-core AVX512-equipped CPU, up to 32,768 residue pairs can be compared at the same time.

Length-based dissimilarity prefiltering. In Prim's algorithm, before computing the exact LLCS between a selected sequence r and each sequence x in R, FAMSA2 estimates a best-case LLCS value assuming that the shorter sequence is a subsequence of the longer one. If this estimate still results in a higher dissimilarity than x's current minimum to M, the exact LLCS calculation is skipped. This heuristic eliminates a few percent of pairwise comparisons, providing a moderate speedup (Supplementary Fig. 1).

Parallel profile–profile alignment. In FAMSA, alignment-stage parallelism was limited to cases where multiple profile pairs could be aligned independently of each other. However, because of the single-linkage guide tree structure, higher tree levels often involve aligning single sequences or small profiles with much larger ones in a sequential, chain-like manner, limiting the parallelism.

FAMSA2 introduces two-level parallelism at the alignment stage. It combines concurrent processing of independent profile–profile alignments (coarse-grained) with multithreaded alignment of a single profile pair (fine-grained), dynamically assigning computational resources between these two levels to keep the CPU busy. The parallelization uses a pool of threads that handles both coarse-grained and fine-grained concurrency and a synchronized queue with profile pairs ready to be aligned (tasks).

As coarse-grained parallelism is more efficient (no thread idling, lack of internal synchronizations), the algorithm prefers it over fine-grained parallelism by assigning every task a number of threads from the pool, expressed as $\max(\lfloor \text{no. of free_threads}/(\text{no. of queued tasks} + 1) \rfloor, 1)$.

Consequently, as long as the no. of queued tasks ≥ the no. of free threads, tasks are assigned with a single thread (coarse-grained parallelism). Otherwise, the number of threads is adjusted to maximize the use of the computational resources. For example, let us assume that there are five tasks in a queue and nine free threads in the pool. The consecutive tasks taken from the queue are assigned with the following number of threads:

(1) → $\max(\lfloor 9/6 \rfloor, 1) = 1$
(2) → $\max(\lfloor 8/5 \rfloor, 1) = 1$
(3) → $\max(\lfloor 7/4 \rfloor, 1) = 1$
(4) → $\max(\lfloor 6/3 \rfloor, 1) = 2$
(5) → $\max(\lfloor 4/2 \rfloor, 1) = 2$

Tasks (4) and (5) are assigned with two threads; thus, they are subject to the fine-grained parallelism. The procedure aligns profiles by calculating a dynamic programming matrix in an anti-diagonal manner in blocks of height = 4 and width = longer profile length/no. of threads assigned. With two threads assigned to the task and blocks grid of height H, the processing scheme requires $H + 1$ iterations and is as follows:

0.	thread 0 → block (0,0),	thread 1 → idle
1.	thread 0 → block (1,0),	thread 1 → block (0,1)
…		
i.	thread 0 → block (i,0),	thread 1 → block (i − 1,1)
…		
H − 1.	thread 0 → block (H − 1,0),	thread 1 → block (H − 2,1)
H.	thread 0 → idle,	thread 1 → block (H − 1,1)

After every iteration, threads are synchronized using actively waiting thread barriers to avoid thread sleep latency. Because of performance reasons, the minimum block width is 256. Therefore, the number of threads assigned to the task is corrected to fulfill this requirement (for example, tasks with longer profile length < 512 are processed by a single thread and tasks with longer profile length < 1,024 are processed by at most three threads).

Dynamic refinement of partial profiles. In FAMSA, refinement was applied only to final profiles with up to 1,000 sequences, as it proved ineffective for larger alignments. In FAMSA2, refinement is used during intermediate steps, specifically when two profiles, each with fewer than 1,000 sequences, are aligned to a profile with more than 1,000 sequences. This ensures refinement occurs while profiles remain small enough for effective improvement, enhancing large-scale alignment quality.

Output alignment compression. Because of FAMSA2's high computational speed, writing alignments to disk can become the primary runtime bottleneck for large datasets. For example, FAMSA2 aligning 3 million sequences (average row length of ~136,000 residues), produced ~408 GB of uncompressed output in 284 s, requiring sustained write speeds exceeding 1.4 GB s^{-1}—well beyond typical HDD performance (~250 MB s^{-1}). FAMSA2 integrates parallel Gzip compression into the output process; alignment data are partitioned across threads, each compressing its blocks independently, allowing compression and writing to proceed concurrently and maintaining overall efficiency.

Built-in column trimming. In phylogenetic analysis, full MSAs are often trimmed to remove poorly conserved columns (uninformative sites), typically using tools such as trimAl[22]. FAMSA2 includes a built-in, simplified version of this step: users can set a minimum nongap fraction per column, which is applied to the output MSA after the alignment, reducing file size and improving downstream analysis efficiency.

Benchmarking alignment accuracy

Tool selection. An initial screening covered 15 MSA methods and parameter variants: Clustal Omega (version 1.2.4; default and two-iteration mode)[12], FAMSA (version 1.1)[13], FAMSA (version 2.4.1; default and medoid tree), Kalign (version 3.4.1)[23], MAGUS (version 0.2.0; default and Clustal Omega guide tree)[24], MAFFT (version 7.526; DPPartTree, PartTree, G-INS-1, and G-INS-1-large)[25], Muscle5 (version 5.3)[10], T-Coffee's regressive method (version 13.46.0.919e8c6b)[11] and UPP (version 4.5.6)[26]. These methods were evaluated on the extHomFam dataset (Supplementary Table 11) to identify tools suitable for large-scale analyses in terms of runtime and memory requirements. On the basis of these criteria, nine methods were selected for downstream benchmarking across multiple datasets and tests: Clustal Omega (default), FAMSA version 1.1, FAMSA version 2.4.1 (default and medoid tree), Kalign, MAFFT with DPPartTree, Muscle5, T-Coffee regressive (using MAFFT-sparsecore as the bottom aligner) and UPP. All runtimes were benchmarked on a workstation equipped with an AMD Epyc 9554 CPU (64 cores clocked at 3.1 GHz) and 1,152 GiB (approximately 1,237 GB) of RAM. Unless otherwise specified, all tools were run using 32 threads. Exact command-line parameters are provided in Supplementary Table 12.

Structure-based benchmarks. The structure-based benchmarks included two complementary datasets. First, we constructed an extended version of the HomFam (version 37.0)[12] (extHomFam) dataset,

following a previous study[13]. The dataset was expanded by incorporating additional protein families from Homstrad[27] (December 2, 2023 release) and sequences from Pfam (version 37.0; UniProt variant)[28], resulting in 390 protein families, each containing between 206 and 2,979,588 sequences (median: 46,855), with 3–41 Homstrad reference sequences (mean = 6) per family. For each family, MSAs were generated using the tested tools and accuracy was evaluated by comparing the aligned Homstrad reference sequences to the original structural alignments. Accuracy was measured using the SP and TC scores (range: 0–100) through T-Coffee's 'aln_compare' option. The SP score reflects the proportion of correctly aligned residue pairs, while the TC score measures the fraction of correctly aligned columns shared with the reference. To assess accuracy with increasing numbers of homologs, we selected 121 extHomFam families containing more than 100,000 sequences. For each family, 20 subsets were generated by incrementally adding between 100 and 100,000 nonreference homologs to the reference sequences, producing alignments of increasing size. Accuracy stability was measured as the difference in SP and TC scores between the first (smallest) and last (largest) subsets. To evaluate the impact of nonhomologous sequences, 20 composite subsets were created from the same 121 extHomFam families. Starting with a subset of 731 reference sequences (that is, reference sequences combined from all 121 families), nonreference homologous sequences were added equally from each family, resulting in subset sizes ranging from 731 to 12,100,731 sequences. MSAs were computed for each subset and accuracy was assessed by calculating SP and TC scores for each family, followed by averaging across all families.

A well-known limitation of the benchmark described above is its reliance on a limited number of reference structures per family, reflecting the uneven experimental coverage of the Protein Data Bank (PDB). To complement this, we constructed an additional structure-based benchmark leveraging the AlphaFold Database Clusters[9], in which all sequences have predicted structures. In these datasets, structural alignments based on AlphaFold models have been shown to achieve accuracy comparable to alignments based on experimental structures. To avoid introducing bias from structure-based alignment methods, we assessed accuracy using the LDDT[14], which estimates structural consistency based on the implied superposition of aligned residues. Accordingly, metadata for 24,963 clusters containing at least 1,000 proteins each were retrieved on March 6, 2025. PDB structures for 119,209,511 proteins were downloaded from the AlphaFold Protein Structure Database[8]. The clusters were divided into 13 size groups, ranging from 1,000 to 679,681 members. For accuracy assessment, 100 clusters were randomly sampled from each size group. For each sampled cluster, an MSA was established and 100 sequences were randomly selected in 50 replicates. Columns containing only gaps were removed to obtain a refined 100-sequence MSA, which, along with the corresponding AlphaFold PDB structures, was used as input for FoldMason (version 7bd21ede)[14] to compute LDDT score, ranging from 0 to 1. The LDDT scores were averaged across the 50 replicates to provide a single accuracy measure for each cluster.

Phylogeny-based benchmarks. Phylogenetic accuracy was evaluated using two complementary benchmarks. They reflect the evolutionary assumptions underpinning MSA, where each column is interpreted as a statement about the shared ancestry of the residues it contains. While structural information is often used to assess alignment quality, it is an indirect proxy and may not fully capture evolutionary correctness[29]. In contrast, phylogenetic assessments aim to evaluate MSAs in terms of their capacity to preserve evolutionary signals. However, because of the lack of reference phylogenies for real data, such assessments rely on either simulations or indirect inference.

First, we constructed a dataset of 1,860 reference MSAs with known phylogenies using the AliSim tool from IQ-TREE (version 2.4.0)[30]. Baseline phylogenies were sampled from random Yule–Harding trees, with branch lengths drawn from exponential distributions defined per simulation setting. Indels were introduced according to per-site insertion and deletion rates, with gap lengths drawn from the distributions specified for each setting. We used 61 distinct evolutionary parameter sets in AliSim, each replicated 30 times, generating a total of 1,860 protein families, varying the number of sequences (1,000–100,000), root-sequence length (400–2,000 residues), branch lengths (0.001–0.999), per-site indel rates (1×10^{-7}–0.006 per site) and substitution models (LG, JTT, WAG) (Supplementary Table 13). Across these families, sequences ranged from 182 to 2,140 residues (mean ± s.d.: 444 ± 220), sequence identity from 8% to 73% (24.7% ± 17.4%) and gap fractions from 0% to 99% (46.8% ± 29.6%) (Supplementary Table 6). Unlike the extHomFam dataset, where alignment accuracy depends on a subset of reference sequences, the SP and TC scores were calculated on the basis of all sequences within each reference MSA. In addition, we computed the expansion score, defined as the ratio of the number of columns in the tool-generated MSA to the number in the corresponding reference MSA, with values closer to 1 indicating better agreement with the reference MSA length (and, typically, higher phylogenetical accuracy).

Second, we aimed to assess whether alignment quality affects the consistency of downstream phylogenetic inference. In practice, MSAs are primarily used for phylogenetic reconstruction. While ML methods are generally considered the most accurate, they are computationally intensive. Simpler distance-based methods, such as ME, provide faster but typically less accurate estimates. However, these two approaches differ in their sensitivity to MSA errors; ML methods consider each alignment column individually, while distance-based methods can average over alignment inaccuracies. Higher-quality alignments should yield greater consistency between ML and ME trees[31]. To test this, we used 121 of the largest families from the extHomFam dataset. Eight families were excluded because of the presence of nonstandard amino acids, which caused failures in phylogenetic tree construction; consequently, 113 protein families were retained for phylogenetic analyses. For each family, we sampled 100,000 sequences and generated alignments using all tested tools. From each alignment, five replicate subalignments (MSA projections) of 100 randomly chosen sequences were extracted, preserving all alignment columns except those containing only gaps.

For each MSA projection, ML trees were inferred using IQ-TREE:

```
iqtree -s <alignment>
```

ME trees were reconstructed using FastME (version 2.1.6.4)[32] with BioNJ distance estimation:

```
fastme -i <alignment>-m BioNJ -p LG -g 1.0 -s -n -z
5 -O <distances>
fastme -i <distances>-o <tree>-s -n -z 5
```

Topological agreement between ML and ME trees was measured using the Robinson–Foulds (RF) distance, which quantifies the number of differing bipartitions between two tree topologies. To obtain a normalized similarity score, the RF distance was divided by the maximum possible RF distance (max_RF) for the given trees. The topological agreement, expressed as a percentage, was then calculated as follows:

$$\text{Topological agreement} = (1 - \text{RF}/\text{max RF}) \times 100\%$$

Agreement values were averaged across replicates and families for each subset size, yielding five topological agreement values per tool.

Function-based benchmarks. In addition to structural and evolutionary constraints, protein sequences encode functional information such as active sites, whose correct alignment is essential for downstream functional annotation. While structural and evolutionary benchmarks may indirectly reflect the alignment of conserved functional sites,

convergent evolution and localized functional constraints may introduce confounding effects. To directly assess alignment accuracy in functional regions, we constructed a dedicated function-based benchmark focused on the conservation of active sites.

The benchmark was built from 772 enzyme domains annotated with 1,376 active sites in Pfam (version 37.1). Protein sequences containing these domains were downloaded from UniProt and active-site residues were predicted using Pfam's 'pfam_scan.pl' tool (version 1.6)[28] with the active site database 'active_site.dat' (version 37.1). For each protein, the sequence corresponding to the annotated domain was extracted from the full-length protein sequence. Domain sequences were excluded if the annotated domain was shorter than 25% of the length of the hidden Markov model domain or if more than 10% of residues were nonstandard amino acids. The resulting domain families contained 1–11 active sites per family (median: 1), ranging in size from 11 to 1,515,446 sequences (median: 29,596), with 0.1% to 100% of sequences having known or predicted active sites (median: 86%).

Alignment accuracy within functional regions was evaluated by computing an active-site entropy for each annotated site. For a given active site, the distribution of -annotated residues across alignment columns was recorded. Let p_i denote the fraction of active-site residues aligned to column i. The entropy was defined as follows:

$$H = -\sum_i (p_i \log_2 p_i),$$

where lower values indicate a higher concentration of active-site residues within a single column (ideal alignment yields $H = 0$). Entropy scores were calculated for all 1,376 active sites and averaged to yield a single accuracy score for each MSA tool.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets generated in this study are publicly available from Zenodo: extHomFam v37.0 dataset (https://doi.org/10.5281/zenodo.6524236)[33], the largest 121 extHomFam families downsampled (https://doi.org/10.5281/zenodo.18418209)[34], the dataset containing 1,166 protein clusters from the AlphaFold Database (https://doi.org/10.5281/zenodo.16082638)[35], the dataset containing 1,860 simulated MSAs with known phylogenies (https://doi.org/10.5281/zenodo.15971352)[36] and the dataset containing 772 enzyme domain families with annotated active sites (https://doi.org/10.5281/zenodo.16023626)[37]. The external reference datasets used to tune the FAMSA2 dissimilarity measure are also available from Zenodo (https://doi.org/10.5281/zenodo.18412132)[38]. Source data are provided with this paper.

Code availability

FAMSA2 is available from GitHub (https://github.com/refresh-bio/FAMSA).

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Author contributions

S.D. and A.G. designed the study and developed FAMSA2. A.G., S.D. and A.Z. performed the benchmarking. A.G., A.Z., C.N. and S.D. analyzed the results and wrote the paper. All authors reviewed and approved the paper.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41587-026-03095-3.

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Extended Data Table 1 | Summary of benchmarking results presented in Fig. 2

Tool	extHomFam (Fig. 2a,b)			Homologous sequences (Fig. 2c)		Homologous and nonhomologous sequences (Fig. 2d)		AlphaFold clusters (Fig. 2e)	Simulated MSA (Fig. 2f)			Tree agreement (Fig. 2g)		Active sites (Fig. 2h)	Average rank
	SP [%]	TC [%]	Time [hours]	SP [%]	TC [%]	SP [%]	TC [%]	LDDT [0-1]	SP [%]	TC [%]	Expansion [0-inf], 1=perfect	SP [%]	Tree agreement [%]	Entropy [0-inf]	
FAMSA2	**79.6**	**61.8**	15.0	**72.1**	**48.1**	**59.7**	**34.9**	**0.510**	**99.39**	**47.5**	0.898	**71.34**	**41.6**	0.149	**1.4**
FAMSA	78.1	60.1	32.0	70.0	46.2	44.9	21.2	0.508	99.37	46.9	0.900	69.13	41.5	0.167	2.6
FAMSA2 medoid tree	76.5	58.5	**0.8**	68.8	44.6	30.9	14.2	0.499	99.25	40.9	**1.048**	69.30	39.7	0.191	3.4
Muscle5	75.7	59.4	315.7	66.9	46.4	–	–	0.431	99.23	37.3	1.610	66.78	40.1	**0.120**	4.6
T-Coffee regressive	78.1	61.2	887.0	68.4	47.6	–	–	0.396	95.67	37.2	6.176	69.92	35.0	0.166	5.1
UPP	70.4	53.6	146.7	62.7	42.2	–	–	0.430	97.50	38.5	1.591	62.10	39.5	0.181	5.4
MAFFT DPPartTree	62.9	44.3	203.0	51.3	28.5	–	–	0.391	91.16	40.6	4.007	50.69	35.0	0.210	6.9
Clustal Omega	63.8	46.9	170.7	50.0	28.5	–	–	0.484	75.93	26.2	2.212	49.44	22.8	0.402	7.0
Kalign	61.9	43.2	1.1	48.6	27.0	1.5	0.1	0.462	89.68	22.2	3.012	48.08	34.8	0.377	7.1

Bold values indicate the top-performing tool in each benchmark. Each tool was evaluated across seven alignment scenarios: large protein families from extHomFam (SP/TC; Fig. 2a,b), increasing-size homologous datasets (SP/TC at largest size; Fig. 2c), mixed homologous and nonhomologous datasets (SP/TC at largest size; Fig. 2d), structural consistency of AlphaFold clusters (LDDT; Fig. 2e), simulated MSAs (SP/TC and expansion; Fig. 2f), guide tree reconstruction (SP/tree agreement; Fig. 2g) and functional conservation of active sites (entropy; Fig. 2h). Accuracy metrics for Fig. 2c,d correspond to the final dataset size. A dash (–) indicates that the tool did not complete the corresponding test. The average rank column shows the mean rank for each method across metrics with numerical results from all methods.

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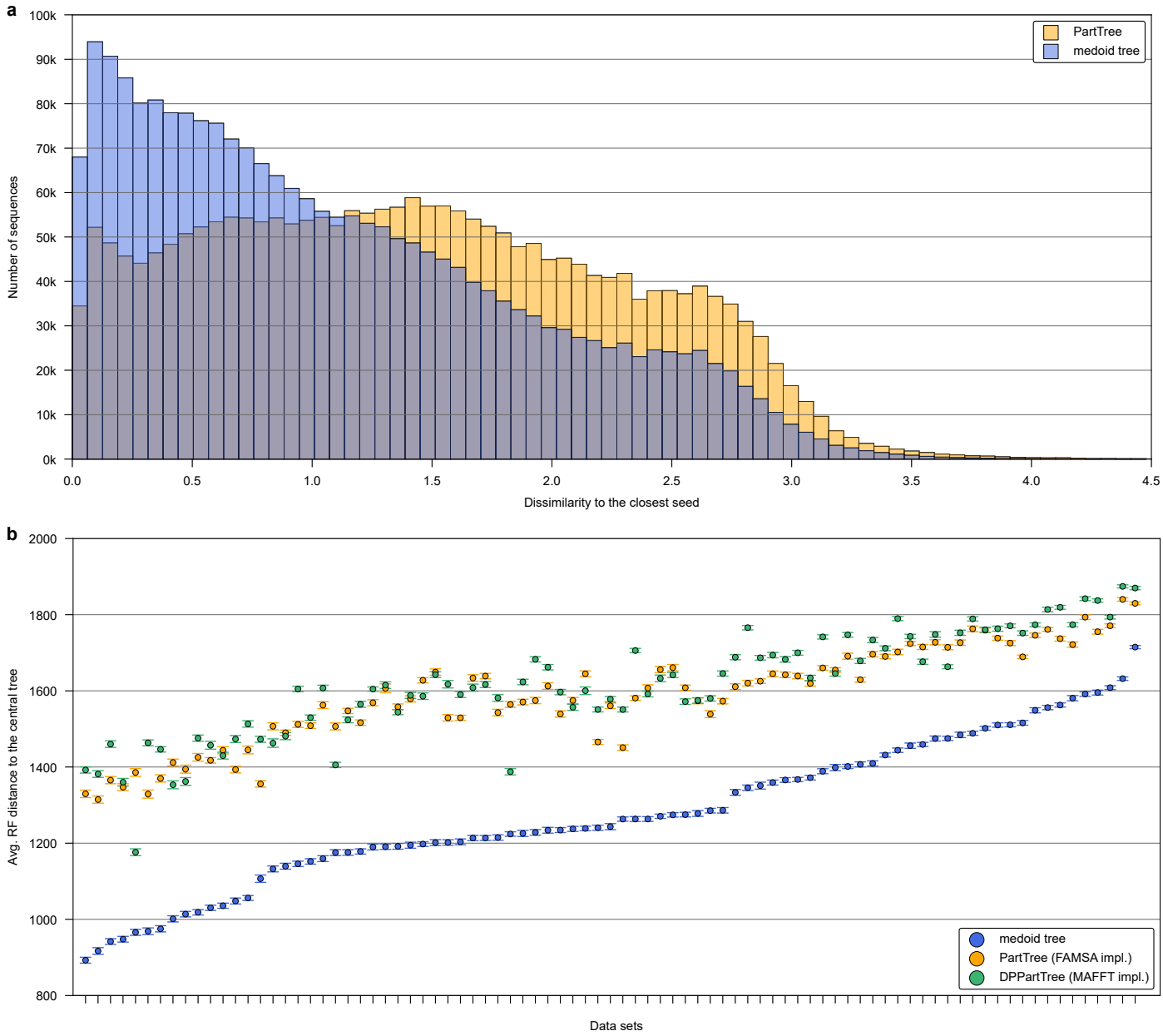

Extended Data Fig. 1 | Properties of medoid trees and its alternative, PartTree from the MAFFT package, evaluated on 85 extHomFam families containing 15,000–40,000 sequences. a, Distributions of dissimilarities between sequences and their closest representatives (seeds) aggregated across all families within benchmark. The visualization concerns only the top recursion level with 50 seeds. Lower dissimilarities to representatives confirm medoid trees to provide better grouping than PartTree (reimplemented in the FAMSA framework to ensure consistency in dissimilarity measure). **b**, Stability of guide trees rendered by medoid tree, FAMSA-implemented PartTree, and MAFFT DPPartTree (PartTree with accurate dynamic programming-based dissimilarity measure). The stability of trees was measured per family (data set) with the following protocol: (i) randomly select 1,000 *core* sequences, (ii) generate 50 replicates, each with 5,000 sequences including all core sequences, (iii) generate a guide tree for every replicate and prune non-core sequences, (iv) calculate Robinson-Foulds (RF) distances between replicate trees, (v) determine central tree as the one minimizing the average RF distance to all other replicate trees (error bars represent 95% confidence interval).

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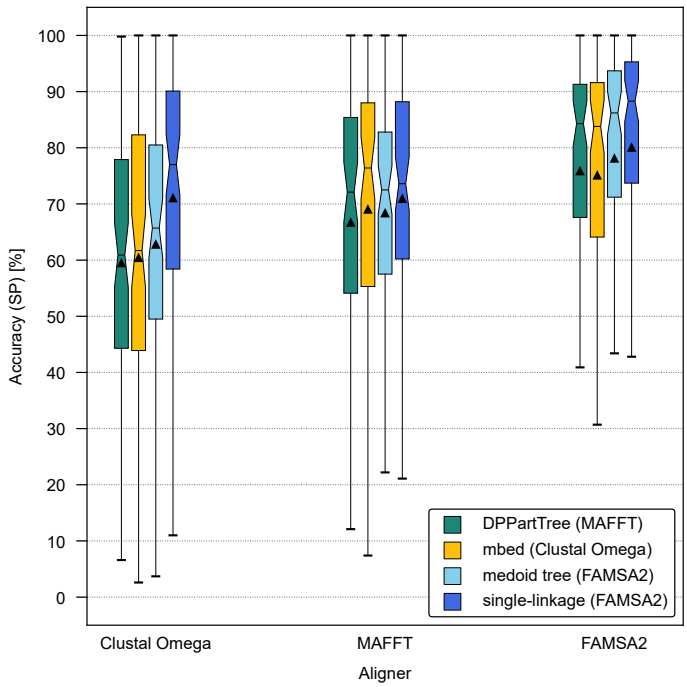

Extended Data Fig. 2 | Accuracy of guide tree and alignment tool combinations evaluated on 85 extHomFam families containing 40,000–100,000 sequences. The plot shows FAMSA2 components to be additively beneficial to the accuracy: FAMSA2 guide trees rendered highest average accuracy independently of the alignment algorithm (Clustal Omega, MAFFT, FAMSA2), and the FAMSA2 aligner rendered highest accuracy independently of the guide tree algorithm (mbed from Clustal Omega, DPPartTree from MAFFT, medoid and single-linkage from FAMSA2). Box plots legend: central line—median, notches—95% confidence interval of the median, triangle—mean, box—lower (Q1) and upper (Q3) quartiles, whiskers—lower (Q1-1.5×IQR) and upper (Q3 + 1.5×IQR) outliers bounds (IQR = Q3-Q1, interquartile range).

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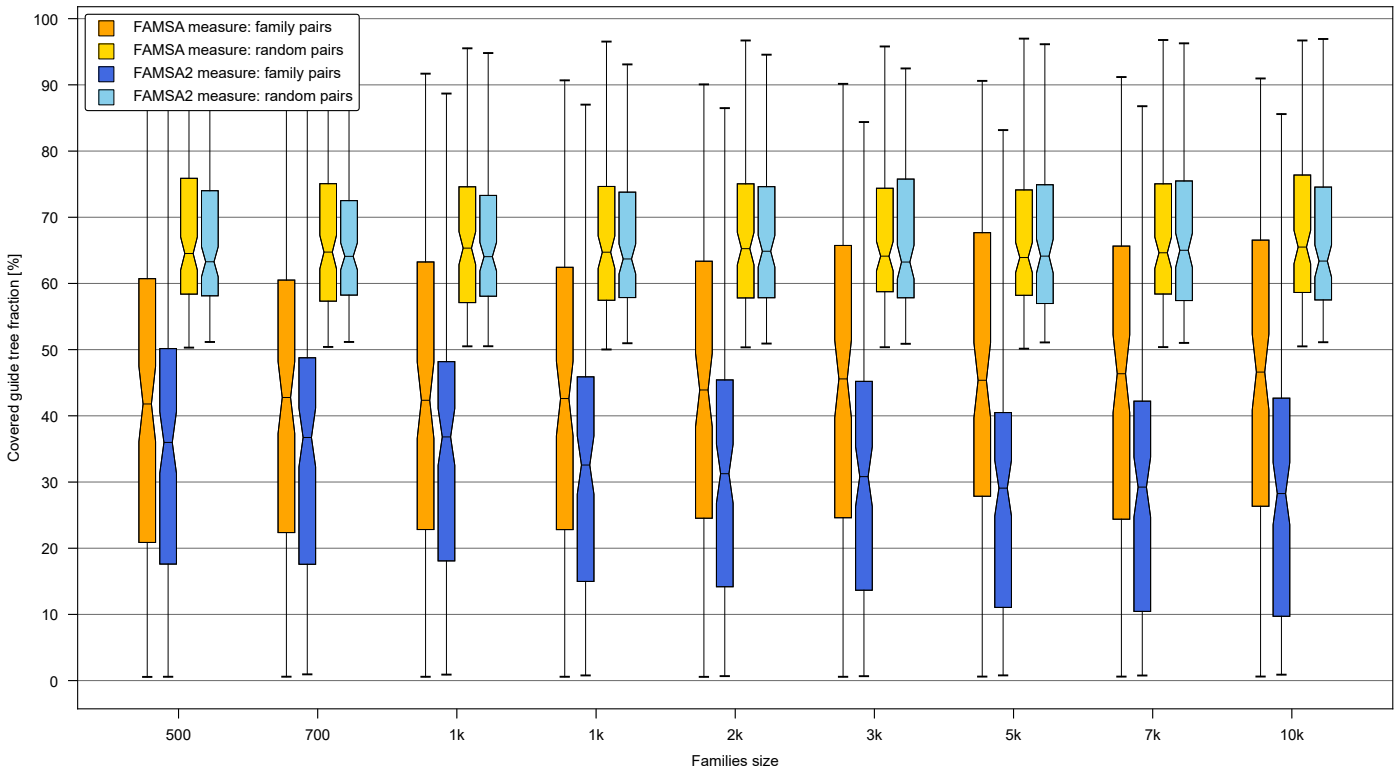

Extended Data Fig. 3 | Comparison of FAMSA and FAMSA2 dissimilarity measures in terms of clustering homologous sequences. Both measures were embedded in a single-linkage guide tree construction algorithm in FAMSA2. Guide trees were generated on concatenated sets of 121 extHomFam protein families with increasing total numbers of sequences. Each guide tree thus contained proteins from all 121 families. To assess the ability to cluster homologous sequences, 10,000 protein pairs were randomly sampled (with replacement) from each family, and the size of the smallest subtree containing each pair (defined by their last common ancestor) was determined. This size was expressed as a percentage of the total number of sequences in the guide tree, indicating how dispersed the pair was within the tree. Percentages were averaged over the 10,000 bootstrap samples per family, and the resulting 121 values (one per family) were visualized as boxplots. Across all dataset sizes, FAMSA2 dissimilarity measure produced significantly smaller subtrees than FAMSA – a series of nine one-sided paired t-tests ($n = 121$) produced p-values ranging from 2.01×10^{-4} (smallest families) to 1.95×10^{-17} (largest families) which, after applying Bonferroni-Holm correction, is significant at level $\alpha = 0.001$. This indicates improved clustering of homologous sequences with FAMSA2. As a negative control, 10,000 random protein pairs were sampled from the entire tree regardless of family, yielding no significant difference between methods (p from 0.506 to 0.541 across different family sizes). Box plots legend: central line–median, notches–95% confidence interval of the median, triangle–mean, box–lower (Q1) and upper (Q3) quartiles, whiskers–lower (Q1-1.5×IQR) and upper (Q3 + 1.5×IQR) outliers bounds (IQR = Q3-Q1, interquartile range).

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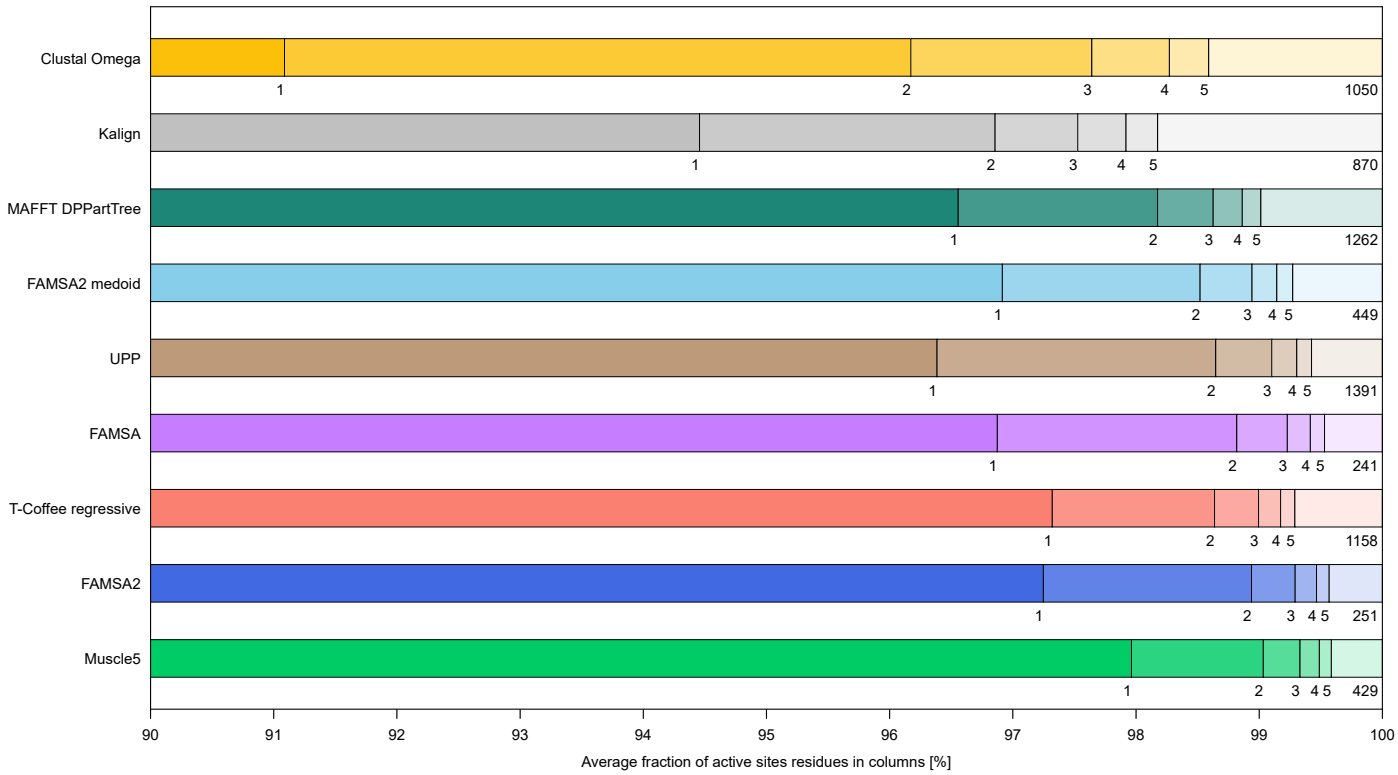

Extended Data Fig. 4 | Concentration of active site residues in alignment columns across 772 Pfam enzyme families. For each active site, we established a distribution of residues across columns, numbered from most to least populated. These distributions were then averaged over all 1376 active sites for each tool and visualized as bars. For clarity, only the five most populated columns (labeled 1–5) are distinguished. The number on the right indicates the maximum number of distinct alignment columns over which an active site was distributed by a particular tool.

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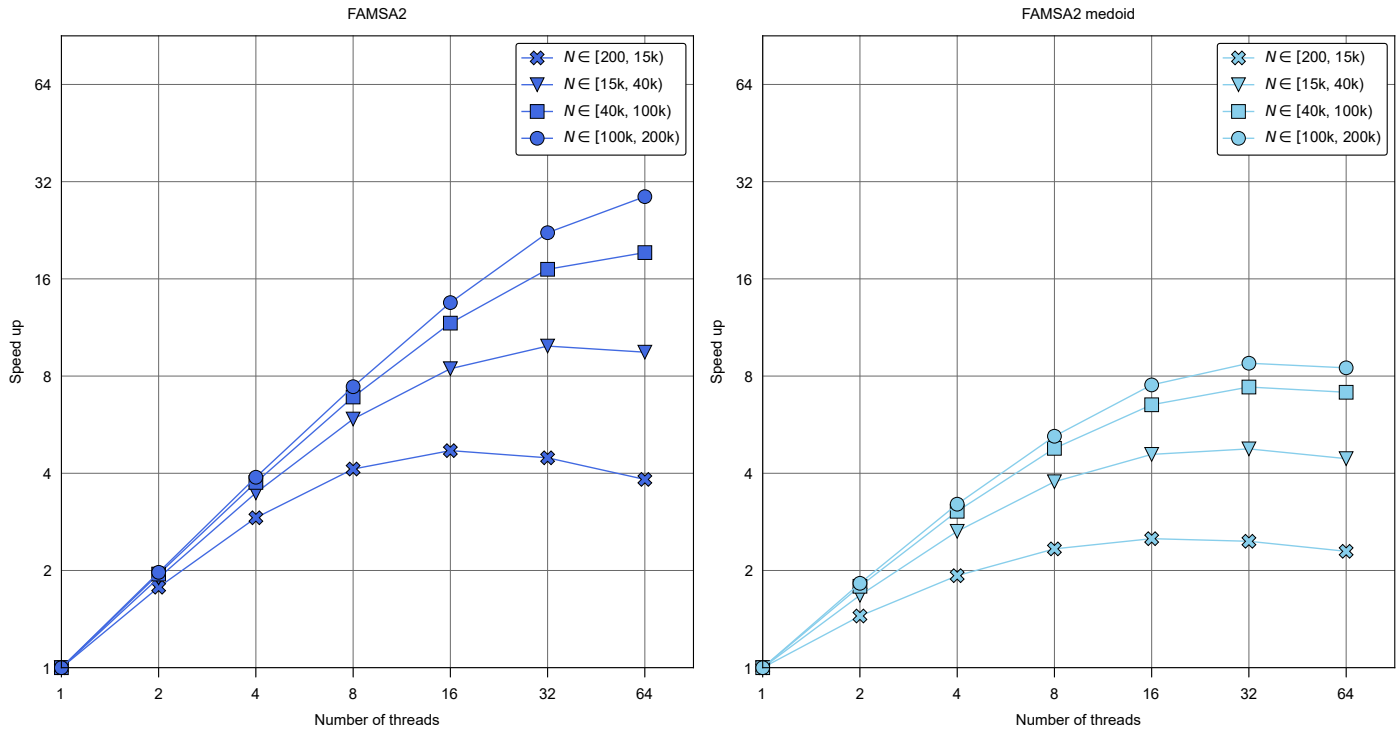

Extended Data Fig. 5 | Scalability of FAMSA2 and FAMSA2 with medoid guide tree with the number of threads on extHomFam protein families. Speedup relative to single-threaded execution is shown for **a**, FAMSA2 and **b**, FAMSA2 medoid as a function of thread count (1–64) on a log-log scale. Analyses were performed on extHomFam protein families grouped by size *N*. FAMSA2 shows near-linear speedup up to 16 threads for larger datasets, achieving over 30-fold acceleration at 64 threads for the largest families, while the medoid tree variant exhibits more limited scalability, saturating around 8-fold speedup. These results demonstrate FAMSA2's efficient multithreading for large-scale protein alignments.

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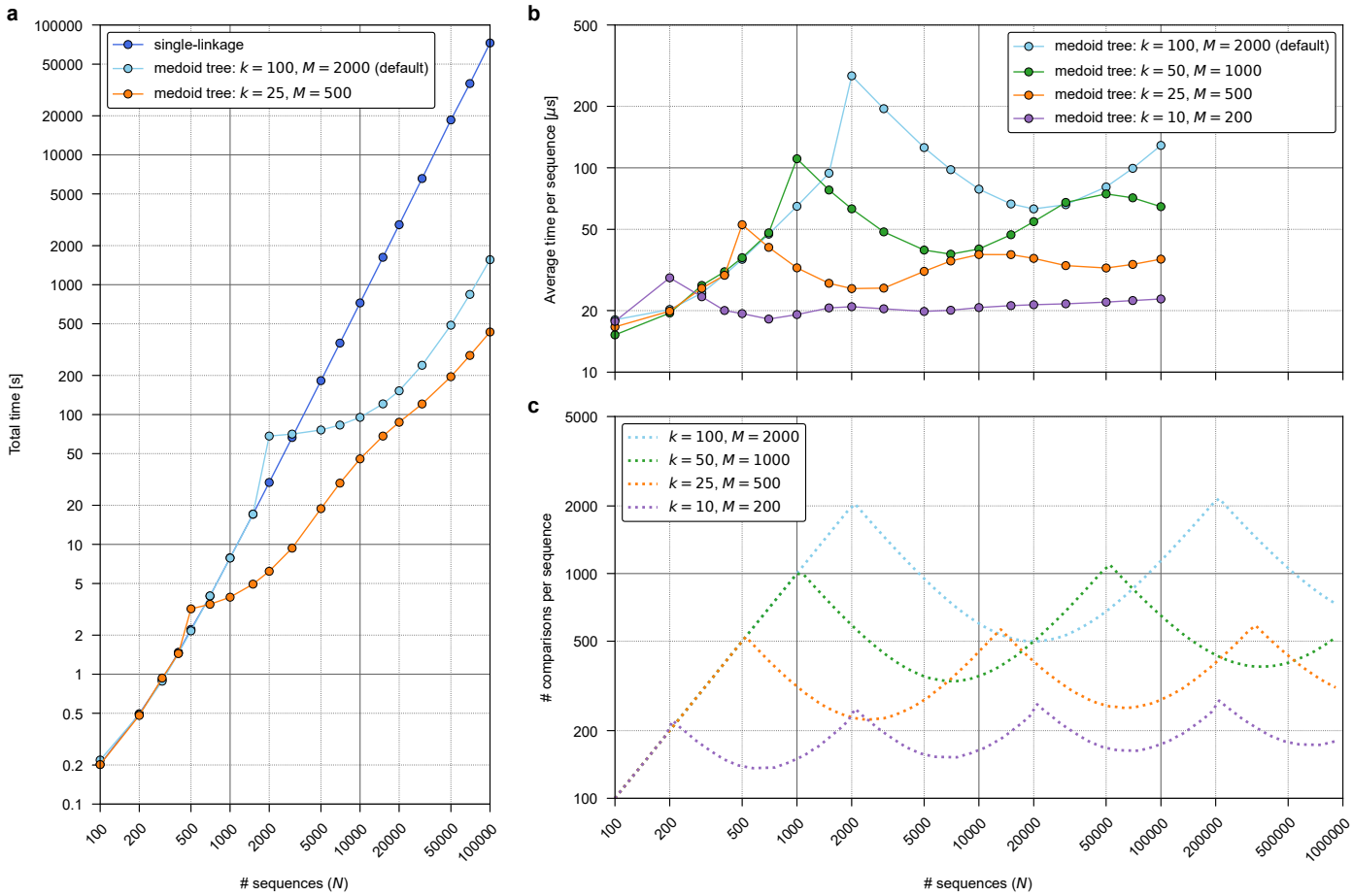

Extended Data Fig. 6 | Scalability of guide tree algorithms in FAMSA2 with the number of input sequences. a, Total time of single-linkage and medoid tree algorithms executed sequentially on the 121 largest extHomFam families subsampled to 100–100,000 sequences. **b**, Average processing time per sequence. **c**, Theoretical number of pairwise comparisons per sequence assuming perfectly balanced splits in the medoid tree algorithm. All panels are shown on log-log scales. The single-linkage trajectory in panel **a** (slope = 2 across all input sizes) is consistent with its $O(N^2)$ time complexity. The medoid tree algorithm follows the same quadratic increase up to the maximum subtree size M, after which the recursion starts and scaling becomes subquadratic. The trajectories of per-sequence processing times in **b** closely follow the log-periodic functions representing the theoretical numbers of sequence comparisons in **c**: (i) peaks at arguments $M \times 10^i, (i = 0,1,2, \ldots)$ reflect altering dominance of the runtime by different algorithm steps (Methods), (ii) an increasing baseline corresponds to the logarithmic growth of the tree height (Methods). Overall, an agreement between empirical results and the simple theoretical model (perfectly balanced splits) confirms that the assumption of $O(MN \log N)$ time complexity of medoid trees (upper-bounded split imbalance; Methods) is met in real-life alignment problems.

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☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used.
Data analysis	Our tool FAMSA v2.4.1 (described in the manuscript and available at https://github.com/refresh-bio/famsa) was used for data analysis. Benchmarking was performed against Clustal Omega v1.2.4, FAMSA v1.1, Kalign v3.4.1, MAGUS v0.2.0, MAFFT v7.526, Muscle5 v5.3, T-Coffee's regressive method v13.46.0.919e8c6b, and UPP v4.5.6. The reference dataset of simulated multiple sequence alignments (MSAs) was generated using AliSim in IQ-TREE v2.4.0. Active site residues of enzyme families were identified using Pfam's pfam_scan.pl tool v1.6 with data from the Pfam v37.1 FTP repository. Structural alignment accuracy was assessed using FoldMason v7bd21ede to compute LDDT scores from MSAs and the corresponding AlphaFold protein structures. Maximum-likelihood phylogenies were inferred using IQ-TREE v2.4.0, and minimum-evolution trees were reconstructed using FastME v2.1.6.4.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The datasets generated in this study are publicly available on Zenodo – the extHomFam v37.0 dataset: https://doi.org/10.5281/zenodo.6524236; the largest 121 extHomFam families downsampled: https://doi.org/10.5281/zenodo.18418209; the dataset containing 1,166 protein clusters from the AlphaFold Database: https://doi.org/10.5281/zenodo.16082638; the dataset containing 1,860 simulated multiple sequence alignments with known phylogenies: https://doi.org/10.5281/zenodo.15971352; the dataset containing 772 enzyme domain families with annotated active sites: https://doi.org/10.5281/zenodo.16023626. The external reference datasets used to tune the FAMSA2 dissimilarity measure are available at https://doi.org/10.5281/zenodo.18412132. Source data are provided with this paper.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	N/A
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The number of datasets was determined by selecting complementary benchmarks that collectively evaluate multiple sequence alignment (MSA) accuracy across four key dimensions: structural correctness (via reference structural alignments and AlphaFold-predicted structures), phylogenetic signal preservation (through large-scale simulations and real-data topological consistency), functional site conservation (active site entropy), and scalability (performance on datasets exceeding 1 million sequences). This multi-faceted approach addresses known limitations of individual benchmarks (such as the sparse experimental coverage of the Protein Data Bank or the indirect nature of evolutionary proxies) ensuring a comprehensive assessment under diverse biological conditions. With thousands of families/clusters and millions of sequences, this study provides robust, generalizable evidence for tool performance that exceeds the scope of prior MSA benchmarks while remaining computationally feasible for state-of-the-art methods. Sample-size calculations were not performed. In most experiments, full datasets were analyzed to capture the complete diversity and scale of the sequences under study. Specifically, the extHomFam and HomFam structural benchmarks included all available families and sequences meeting inclusion criteria, and the simulation-based phylogeny benchmarks used all 1,860 MSAs generated across 61 evolutionary parameter sets with 30 replicates each. Exceptions where full datasets were not used were due to computational constraints: in the AlphaFold-based structural benchmark, 100 clusters were randomly sampled from each of 13 cluster-size bins, and for each sampled cluster, 100 sequences were randomly selected in 50 replicates; in phylogeny-based analyses of the largest real-data families, five replicate sub-alignments of 100 sequences were extracted from each full alignment to allow maximum-likelihood and minimum-evolution tree inference. Function-based benchmarks included all enzyme families passing quality filters, with no subsampling required. In all cases, replicate sampling and averaging were applied to ensure that results were robust to stochastic variation and representative of the underlying datasets.
Data exclusions	No data were excluded from the majority of analyses. Exceptions included eight protein families from the extHomFam dataset that contained nonstandard amino acids, which caused failures in phylogenetic tree construction; these families were excluded from phylogeny-based benchmarks, and the exclusion criterion was pre-established based on sequence validity. Additionally, domain sequences in the function-

based benchmark were excluded if the annotated domain was shorter than 25% of the corresponding HMM model or contained more than 10% non-standard amino acids; these criteria were applied prior to analysis to ensure reliable functional site assessment. All other datasets were analyzed in full, and all inclusion and exclusion decisions were applied consistently across tools and replicates.

Replication	Testing was conducted on various machine configurations, including Linux and Windows x64-based desktops, Linux x64-based servers, MacOS x64-based desktops, and MacOS ARM-based desktops. The results were consistent across all configurations.
Randomization	Randomization in the traditional experimental sense was not applicable, as all analyses were performed on pre-existing protein sequence datasets rather than experimentally assigned groups. Where subsampling was necessary (i.e., selecting sequences from large clusters in the AlphaFold structural benchmark and generating replicate sub-alignments for phylogeny-based analyses) sequences were sampled uniformly at random within clusters or alignments to avoid bias and preserve the underlying sequence diversity. Stratification by cluster size or family identity ensured that samples remained representative of the full dataset. All other analyses used the entire available datasets, maintaining complete representation of sequence diversity.
Blinding	Not applicable, as this paper focuses on method development and benchmarking.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A